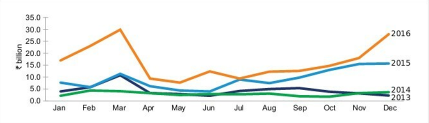


*Fig. 4: Solar module imports by India (Source: Ministry of Commerce, GoI)*

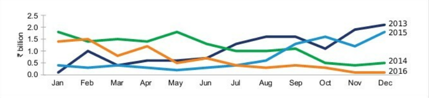


*Fig. 5: Solar module exports from India (Source: Ministry of Commerce, GoI)*

Though solar module imports have increased significantly in the last four years in line with growth in capacity addition, exports have dropped notably (Fig. 5).

This increasing dependence on imports in a growing and strategically important solar sector has created various stress points in the Indian economy, which may lead to an abrupt policy reaction by the government. However, the Indian government needs to consider long-term implications for the sector and chalk out a well thought-out plan for domestic manufacturing instead of introducing short-term support measures to counter China's monopoly in this sector.

China has announced huge subsidies and other support measures to scale up its solar PV manufacturing and dominate the global market. This has resulted in a massive increase in the Chinese manufacturing capacity from 23GW in 2013 to over 70GW today, despite the steep fall in prices. China also provides support for new technologies through its 'Top Runner' programme, which encourages the industry to migrate to higher-efficiency PV products.

So far, the Indian government's priority has been to increase generation capacity and lower tariffs. But in the absence of a proper domestic manufacturing ecosystem, such measures are not enough to enable Indian manufacturers to compete with Chinese imports.

## Is GST 'good and simple' enough?

In accordance with the 'one nation one tax' policy, solar panels and solar PV cells now attract a concessional GST rate of 5 per cent. This GST rate as against the effective 0 per cent rate of taxation will increase the total project costs.

However, the increase in project costs may be offset by the following factors:

1. Removal of the cascading effect of taxes
2. Availability of input credit resulting in improved cash flows and EBITDA (Earnings Before Interest, Taxes, Depreciation and Amortisation)
3. Reduction in administrative and warehouse costs due to the abolition of multiple tax compliance requirements
4. Reduction of tariff-related trade barriers across the country enabling the growth of a common market

The same GST rate of 5 per cent has been extended to all equipment for solar power generating plants or other capital goods used in solar projects. However, there is still a lack of clarity on 'other' capital goods used in solar projects. There is confusion in the market on how to avail the concessional 5 per cent GST rate when many of the same components are taxed at higher rates for use in other industries. For example, transformers are taxed at 5 per cent for solar projects as against 18 per cent for other applications. In the absence of any specific notification for solar projects, the GST rates vary from 18 per cent for capital goods, such as inverters and module mounting structures, to 28 per cent for cables and batteries.

According to BTI estimates, if all capital goods for solar projects are taxed at 5 per cent rate, the overall increase in EPC costs will be around 3 per cent. However, if only modules are taxed at 5 per cent and other capital goods are taxed at 18-28 per cent rates, the net increase in EPC costs is expected to be around 6 per cent.

With global solar module prices firming up due to extension of the solar Feed-in Tariff deadline in China, GST has come at an unfavourable time for the solar sector in India. Project delays can be expected and one cannot rule out the risk of litigation between project developers and discoms over the sharing of additional costs. The adverse impact will also be felt in the rooftop solar market, where market activity is expected to slow down in the near future.

## Moving forward

Solar power as a clean energy technology option has already surpassed expectations, and this trend could continue. However, availability of grid-scale energy storage is critical to ensure sustainable growth of the solar sector. Therefore the onus lies with the energy storage industry to deliver low-cost yet reliable storage solutions. EFY